CSE-4632

Digital Signal Processing Lab

Lab 2

Lab Group: **1**

Thursday, 21th May, 2026

Task 1:

Background.

A continuous-time signal

$$x(t) = A\sin(2\pi f_0 t)$$

is sampled at rate f_s to obtain the discrete-time sequence

$$x[n] = x(nT_s)$$

where $T_s = 1/f_s$.

Tasks:

1. Generate the analogue signal

$$x(t) = \sin(2\pi \cdot 5t)$$

over the interval $t \in [0, 1]$.

2. Sample the signal at:

$$f_s \in \{8,\ 5,\ 3\}\ \text{Hz}$$

3. Reconstruct the sampled signal using Zero-Order Hold (ZOH).

4. Plot:

- original signal,
- sampled points,
- reconstructed signal.

Task 2:

Background.

An N-bit uniform quantiser divides the amplitude range into 2^N levels.

Tasks:

1. Generate:

$$x(t) = \sin(2\pi \cdot 4t)$$

sampled at $f_s = 100$ Hz.

2. Quantise the signal using:

$$B \in \{2,\ 4,\ 8\}$$

bits.

3. Compute the quantisation error:

$$e[n] = x_q[n] - x[n]$$

4. Plot the quantisation error.

5. Compute the measured SQNR and compare it with:

$$\text{SQNR} \approx 1.76 + 6.02B$$

Task 3:

$$x[n] = \begin{cases} n^2 - 1, & -2 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Tasks:

1. Generate the signal values.
2. Create a tabular representation.
3. Write the sequence representation.
4. Plot the signal using a stem plot.

Task 4:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Tasks:

1. Generate and plot $\delta[n]$.

2. Plot:

$$3\delta[n-2]$$

and

$$-2\delta[n+3]$$

on the same graph.

3. Construct the unit step using impulses:

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

4. Verify the result graphically.

Task 5:

Tasks:

1. Define:

$$x[n] = \{1, 2, 3, 2, 1\}$$

2. Compute:

$$y_1[n] = x[n-3]$$

and

$$y_2[n] = x[n+2]$$

3. Plot all signals.

Task 6:

$$x[n] = \{3,\ 1,\ -2,\ 4,\ 2\}$$

Tasks:

1. Decompose the signal into shifted impulses.
2. Plot each impulse component separately.
3. Sum all components.
4. Verify the reconstructed signal.

Task 7:

$$r[n] = n\,u[n]$$

Tasks:

1. Generate and plot $r[n]$.
2. Verify:
$$u[n-1] = r[n] - r[n-1]$$
3. Plot:
$$r[n-2]$$
and
$$0.5r[n]$$
4. Compare the resulting signals.